UAS SafeNet Luna Payment HSM Deployment Guide - 6.0.1-GA (AI)

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1. Preface - UAS SafeNet Luna Payment HSM Deployment Guide

This guide contains instructions on the installation and set up of SafeNet Luna Payment Hardware Security Module (HSM) formerly known as Luna Electronic Fund Transfer (Luna EFT) HSM.

Intended Reader

This document is intended as reference for administrators.

Document Scope

This guide covers the first-time setup and configuration of the SafeNet Luna Payment HSM.

You may refer to SafeNet's Luna Payment HSM documentation for the complete installation steps and details regarding the configuration and setup of the HSM.

For any feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix UAS (Universal Authentication Server) documentation set:

- *AccessMatrix Release Notes*
- *AccessMatrix Common Administration Guide*
- *UAS Administration Guide*

2. Introduction to UAS SafeNet Luna Payment HSM Deployment

The basic functions of the SafeNet Luna Payment HSM are to manage and secure the key storage. There are two types of keys that can be managed via the SafeNet Luna Payment HSM:

- Symmetric keys e.g. the KTPV (Key for Transformed Pin Value)
- Asymmetric keys e.g. the RSA public key used in the E2EE java applet

Symmetric Keys are entered via the following method:

1. First, X numbers of key custodians are assigned for each key.
2. Each of the key custodians selects a random component from the SafeNet Luna Payment HSM random component generator and stores it in a key slip.
3. All the key custodians then enter their respective key components for those specific keys into the SafeNet Luna Payment HSM. The key components are automatically combined and stored into the SafeNet Luna Payment HSM.
4. All the key custodians then seal up their key slips and store them in the respective safes.

RSA keys are generated in the SafeNet Luna Payment HSM. The user specifies the key index, RSA Key length and the modulus via the SafeNet Luna Payment HSM console. Thereafter, the SafeNet Luna Payment HSM automatically generate the RSA keys.

The SafeNet Luna Payment HSM also provides pin generation, printing and verification functions. When a pin generation instruction is sent to the HSM, it randomly generates a password/pin according to a configurable password policy within the HSM. It connects to a serial printer via a serial port so that pin mailer can be printed when a new password is generated. Upon successful generation of new password, an encrypted password called the TPV (Transformed Pin Value encrypted by KTPV) will be returned to the caller so that it can be used for password verification at a later stage. If remote printing is used, the PIN Protect Key (PPK) will be used to encrypt the password before sending the password to the HSM for remote printing. For detailed information on E2EE processes, please refer to the E2EE documentation.

SafeNet Luna Payment HSM can also provide additional protection for application's passwords, e.g. database password, AccessMatrix passwords, by offering encryption and decryption using Data Protect Key (DPK) and Domain Master Key (KM). DPK is used to encrypt and decrypt application's passwords, whereas KM is used to encrypt HSM keys, such as DPK.

The HSM communication is via TCP/IP. It listens to a configurable TCP port number. Access control is based on IP address. To ensure security, only IP addresses stored in the host address list of the HSM can establish a connection with the HSM.

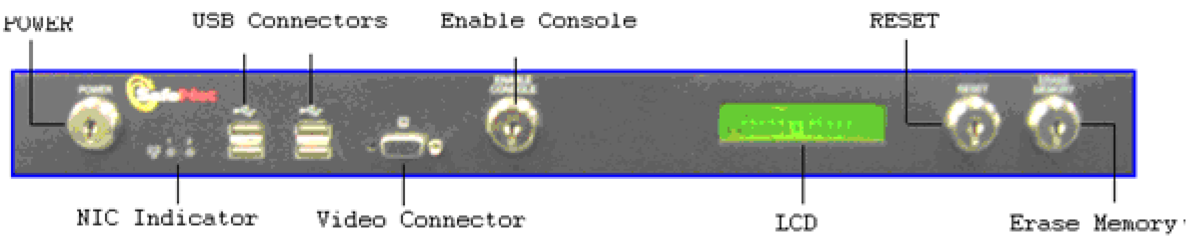
For better reliability, SafeNet Luna Payment HSM supports the backing up and restoring of data, and also failover load balancing with the help of the Security Module Configuration in the AccessMatrix server. In order to support failover load balancing, the HSM used for load balancing must have the same RSA keys and KTPV. This can be done by transferring (backing up and restoring) the keys from one HSM to another using a set of smartcards. For data protection in the smartcard, the data is encrypted using the Smartcard Transport Protect Key (*KTP) and further secured using smartcard ID and pin.

Keys

SafeNet Luna Payment HSM has four kind of keys used to operate: 2 x Power/Reset key (blue color-coded), Console key (green color-coded), Erase Memory (red color-coded)

!	**Important Note**: Do not open the chassis without permission. Opening the chassis will erase all data.

Front Panel

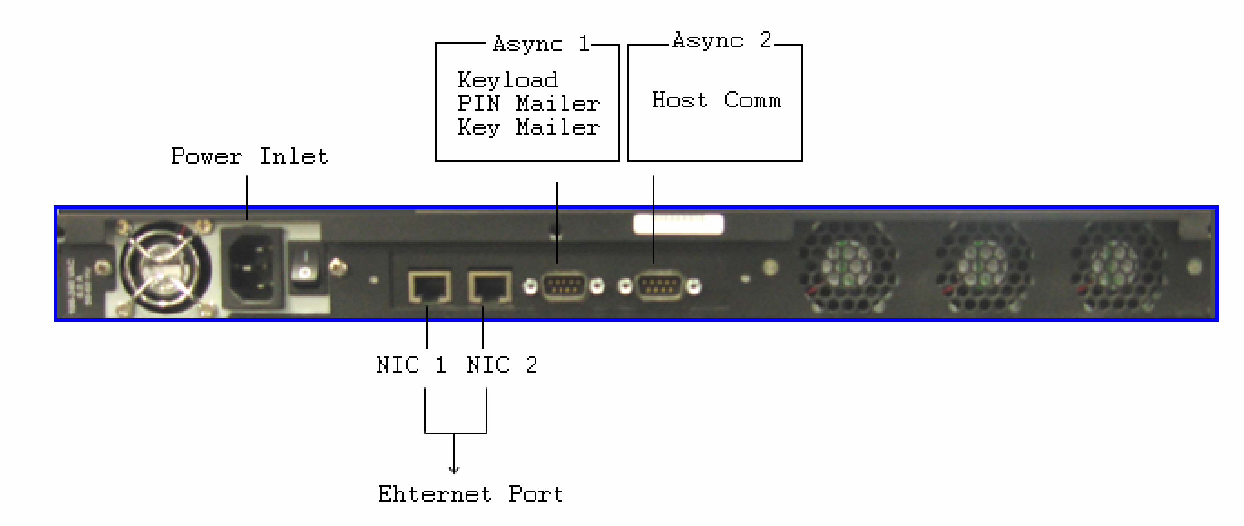


The front interface has eight components:

- **Power Keyhole** - Insert the power key and turning the key clockwise to ON to turn on the HSM, turning the key counter-clockwise to turn off the HSM. (HSM might take some time to stop the processes and shut down.)
- **NIC Indicator** - The NIC indicator the network status. 2 NIC indicators are present on the front panel. At present, only NIC 1 indicator is functional and flashes green to indicate that NIC1 is functional.

- **USB Connector** - There are 4 USB connectors (enabled via the console switch) present on the front panel designated for keyboard, mouse, smartcard reader (omni key 3121 is supported) and USB CD-ROM or pen drive.
- **Video Connectors** - The 15-PIN VGA Connector enabled via the console switch provides a VGA video output for an analog VGA monitor.
- **Enable Console Keyhole** - Console operation are not possible until the console has been enabled. To enable the console, insert the Enable Console Key into the Enable Console switch and turn it clockwise until you meet resistance.
- **LCD** - The LCD indicates if the Power is On or Off and show various information of the SafeNet Luna Payment HSM. If a Low Battery indication is displayed on the LCD, backup all your keys and contact SafeNet Customer Support.
- **Reset Keyhole** - Use the RESET facility to RESTART the HSM. Insert the key from the Power Keyhole and turning the key clockwise; hold it for 5 seconds to restart the HSM.
- **Erase Memory Keyhole** - To erase the memory of the HSM. Any information that has been entered to the HSM will be destroyed.

Rear Panel



The rear interface has three components:

- **Power Inlet** - An inlet is provided for a 100-240 V 50-60 Hz power.
- **Ethernet Port (Network Interface Card)** - SafeNet Luna Payment HSM has two Ethernet ports (NIC 1 and NIC 2) and only one port is functional. Ethernet port 1 (NIC 1) flashes green when NIC 1 is functional.

- **Asynchronous Port** - SafeNet Luna Payment HSM has two asynchronous ports. Asynchronous Port 1 is used for Keyload, PIN Mailer and Key Mailer, and Asynchronous Port 2 is reserved for host communication.

3. Configuring SafeNet Luna Payment HSM

Starting SafeNet Luna Payment HSM for the First Time/ after Memory Erasure

HSM basic configuration (steps 1 to 8)

1. From the **Console Task Selection** menu, choose the **Main Task Selection** menu and press Enter, you will be asked to enter a new password for Main Task Selection.
2. From the **Main Task Selection** menu, choose the **Operations** menu and press Enter. The **Operations** menu appears.
3. From **Operations** menu, choose the **Smartcard Key Transfer** menu and press Enter, you will be asked to enter a new password for Smartcard Key Transfer.
4. Press Escape several times until **Console Task Selection** menu appears.
5. From the **Console Task Selection** menu, choose the **OBM Pin Mailer** menu and press Enter, you will be asked to enter a new password for OBM Pin Mailer.
6. Set the date and time. Please refer to the Setting Date and Time section.
7. Configure the network setting and access control. Please refer to the Network Setting and Access Control section.
8. Configure the Security Module Host Comm in the AccessMatrix Server; please refer to the Configuring Security Module Host Comms section. If failover load balancing is required, then please refer to the Failover Load Balancing section.

Additional steps for password generation (steps 9 to 18)

9. Store the smartcard Transfer Protect Key. If you would like to restore keys from smartcard in the next step, please use the same Transfer Protect Key used to backup to the smartcard. Please refer to the SmartCard Transfer Protect Key (*KTP) section.
10. Restore keys from smartcard, if necessary. Please refer to the Retrieving or Restoring Keys from SmartCard section. If you restore keys from smartcard, you may want to skip step 10 and step 13.
11. Configure RSA. Please refer to the Managing HSM RSA Key Length section.
12. Store the Domain Master Key (KM). Please refer to the Domain Master Keys (*KM) section.
13. Store the Transformed PIN Value Key (KTPV). Please refer to the Creating Transformed PIN Value Key (KTPV) section.

14. Store the Data Protect Key (DPK), if required. Please refer to the Creating Data Protect Key (DPK) section.
15. Store the PIN Protect Key (PPK). Please refer to the Creating PIN Protect Key (PPK) section.
16. Set password restrictions. Please refer to the Setting Password Restriction section.
17. Enable transferable HSM RSA keys. Please refer to the Enabling Transferable HSM RSA keys section.
18. Enable OBM Password Mailer Switch to activate the HSM functions properly. Please refer to the Enabling or Disabling OBM Password Mailer Switch section.

Additional steps for password printing (steps 19 to 23)

19. Store PPK if you have not done so in step 13. Please refer to the Creating PIN Protect Key (PPK) section.
20. Configure printer port. Please refer to the Configuring Printer Port section.
21. Align printer. Please refer to the Printing Alignment (Printing Parameters) section.
22. Set printer control string. Please refer to the Printer Control Strings Setting section.
23. Limit Enable Password Mailer, if necessary. Please refer to the Enabling Password Mailer section.

3.1. Configuring Luna Payment HSM

This section describes how to configure the SafeNet Luna Payment HSM, such as configuring network access and changing access control password.

Changing Main Task Selection Access Control Password

Main Task Selection access control password only applies for accessing **Main Task Selection** menu. In order to change the Main Task Selection access control password, please do the following steps:

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **PHEFT administration** and press **Enter**. The **PHEFT administration** menu appears.
3. From the **PHEFT administration** menu, choose **Change Access Control password** and press **Enter**. The **Change password** dialog appears.
4. Enter the old password, new password, and confirm new password.

   > **Notes**:
   >
   > - The password can be made of any ASCII characters.
   > - The length of the password must between six to eight characters.
   > - Password is case-sensitive.
   > - The new password and confirm new password must be identical.

5. Select the **Enter** button to save the password change.

Changing OBM PIN Mailer Access Control Password

OBM PIN Mailer access control password only applies for accessing OBM Pin Mailer menu. In order to change the OBM PIN Mailer access control password, please do the following steps:

1. From the **Console Task Selection** menu, choose **OBM Pin Mailer** and press **Enter**. The **OBM Pin Mailer** menu appears.
2. From the **OBM Pin Mailer** menu, choose **OBM PIN mailer access password update** and press **Enter**. The **Change password** dialog appears.
3. Enter the old password, new password, and confirm new password.

> **Notes**:
>
> - The password can be made of any ASCII characters.
> - The length of the password must between six to eight characters.
> - Password is case-sensitive.
> - The new password and confirm new password must be identical.

4. Select the **Enter** button to save the password change.

Changing SmartCard Access Control Password

SmartCard access control password only applies for accessing Smartcard Key Transfer menu. In order to change the SmartCard access control password, please do the following steps:

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Operations** and press **Enter**. The **Operations** menu appears.
3. From the **Operations** menu, choose **Smartcard Key Transfer** and press **Enter**. The **Smartcard Key Transfer** menu appears.
4. From the **Smartcard Key Transfer** menu, choose **Change Smartcard access control password** and press **Enter**. The **SmartCard Access Change password** dialog appears.
5. Enter the old password, new password, and confirm new password.

> **Notes**:
> - The password can be made of any ASCII characters.
> - The password can be made of any ASCII characters.
> - The length of the password must between six to eight characters.
> - Password is case-sensitive.
> - The new password and confirm new password must be identical.

6. Select the **Enter** button to save the password change.

Setting Date and Time

For setting date and time of the SafeNet Luna Payment HSM, please do the following steps:

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.

2. From the **Main Task Selection** menu, choose **PHEFT administration** and press **Enter**. The **Operations** menu appears.
3. From the **PHEFT administration** menu, choose **Date and time** and press **Enter**. The **Date and time** dialog appears.
4. Enter the date and time accordingly.

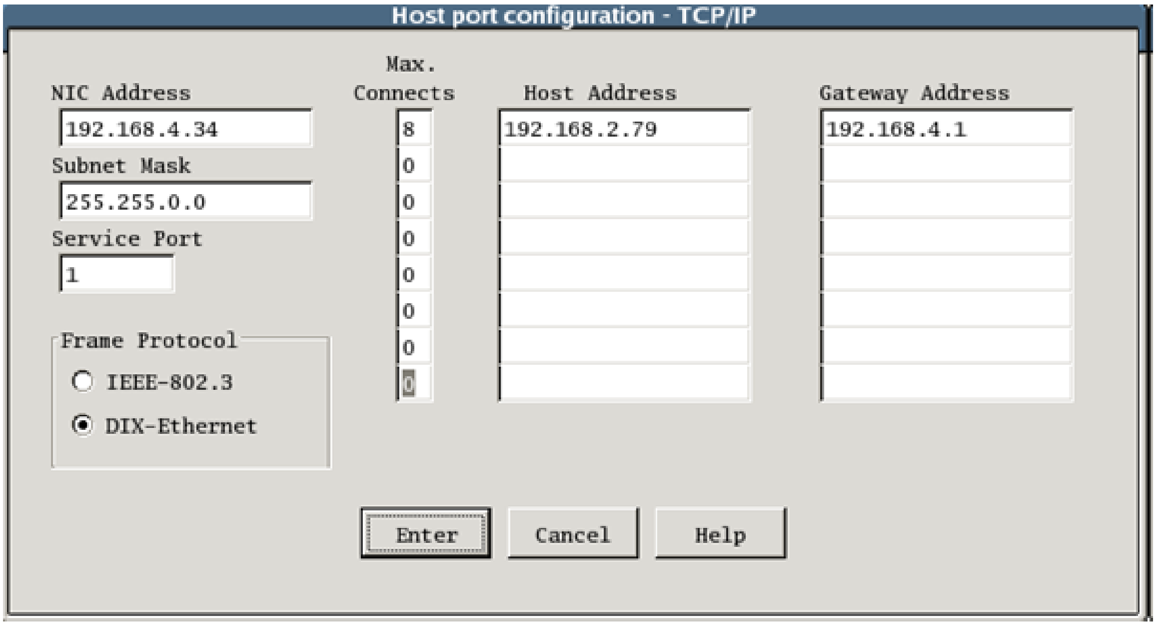
> **Notes**:
> - Date has a valid range of 01-01-1980 to 31-12-2079.
> - Time has a valid range of 00:00 to 23:59.

5. Select the **SAVE** button to save the new date and time.

Network Setting and Access Control

In order for other hosts to access the SafeNet Luna Payment HSM, you have to configure the HSM network and network access control. Only IP addresses stored in the host address list can access the HSM.

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **PHEFT administration** and press **Enter**. The **PHEFT administration** menu appears.
3. From the **PHEFT administration** menu, choose **Configure host communications** and press **Enter**. The **Host port configuration - TCP/IP** dialog appears, as shown below:



4. Enter the following information:
    - IP address of HSM in the **NIC Address** field.
    - Subnet masks information in the **Subnet Mask** field.
    - Service port of HSM in the **Service Port** field.
5. Select the frame protocol used:
    - **IEEE-802.3**, or
    - **DIX-Ethernet**
6. For access control, enter the IP address of the hosts that are allowed to access the HSM in **Host Address**.
7. Enter the maximum number of connection for that particular host in the **Max. Connects** of the respective host.
8. Enter the **Gateway Address** for communication to the respective host, if necessary.
9. Select the **Enter** button.

Managing HSM RSA Key Length

On boot up, 16 RSA key pairs are generated automatically inside the SafeNet Luna Payment HSM. These keys are used for the supported AS2805 processing of the HSM.

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Operations** and press **Enter**. The **Operations** menu appears.
3. From the **Operations** menu, choose **HSM RSA key length management** and press **Enter**. The **HSM RSA key length management** menu appears.
4. From the **HSM RSA key length management** menu, choose **Enter HSM key length** and press **Enter**. The **Enter key length for HSM keys** dialog appears.
5. Enter the length of the key for each index accordingly.

> **Notes**:
> - The length of the key must be between 512 and 2048 (increments of 64 bits).
> - The default key length for these keys is 1024 bits.

6. Select the **Enter** button to save the information.

Managing Host-Stored RSA Key Type

The SafeNet Luna Payment HSM has a table that stores a maximum of 16 key-pairs that are pre-generated on request. Once a key pair is retrieved, it is deleted from the table and a new one is generated.

If a requested key type is not in the table, a key of that type is added to the table and generations starts. If the table is full, an error Public Key Pair not available is returned. If a requested key type is in the table but all keys of that type have been used, an error Public Key Pair generating is returned.

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Operations** and press **Enter**. The **Operations** menu appears.
3. From the **Operations** menu, choose **Host-stored RSA key type management** and press **Enter**. The **Host-stored RSA key type management** menu appears.
4. From the **Host-stored RSA key type management** menu, choose **Enter Host-stored key type** and press **Enter**. The **Key type entry for Host-stored keys** dialog appears.
5. Enter the number of keys, key length, and public exponent, accordingly.

   > **Note**: The key length must be between 512 and 2048 (increments of 64 bits)

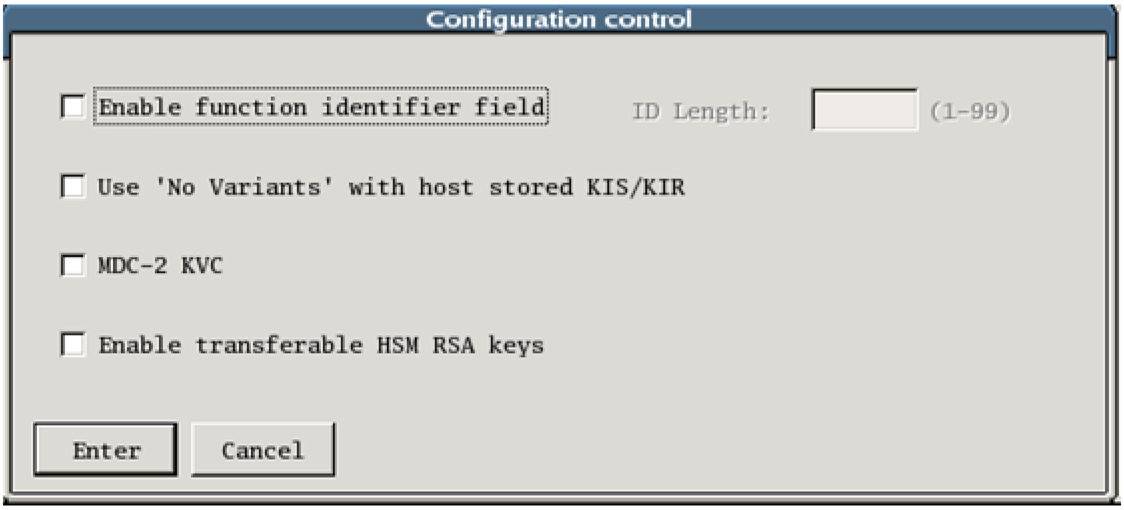
6. Select the **Enter** button to save the information.

Enabling Transferable HSM RSA keys

In order to configure the SafeNet Luna Payment HSM to transfer the HSM RSA keys to smartcard in backing up operation, please do the following steps:

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **PHEFT administration** and press **Enter**. The **PHEFT administration** menu appears.

3. From the **PHEFT administration** menu, choose **Configuration control** and press **Enter**. The **Configuration control** dialog appears, as shown below:



4. Enable the **Enable transferable HSM RSA keys** option.
5. Click the **Enter** button to save the configuration.

Enabling or Disabling OBM Password Mailer Switch

OBM Password Mailer Switch must be enabled or turned on in order for HSM function to work properly. If the HSM is used for printing, this setting is closely related to the Enabling Password Mailer section.

> **Note**: After the limit specified in Enabling Password Mailer has been exceeded, then the HSM will automatically turn off or disable the **OBM Password Mailer Switch**.

1. From the **Console Task Selection** menu, choose **OBM Pin Mailer** and press **Enter**. The **OBM Pin Mailer** menu appears.
2. From the **OBM Pin Mailer** menu, choose **OBM password mailer** and press **Enter**. The **OBM password mailer** menu appears.
3. From the **OBM password mailer** menu, choose **OBM password mailer enable switch** and press **Enter**. The **OBM password mailer enable switch** dialog appears.
4. Select **Off** or **On** accordingly.
5. Select **Enter** button to save the configuration.

Configuring Security Module Host Comms

Security Module Host Comms Configuration utility is used to administer the communication between AccessMatrix and Security Module like HSM. The connection information of HSM like IP address and port number are required for this configuration. The utility can also be used to set up Failover Load Balancing, which will be explain in Failover Load Balancing section.

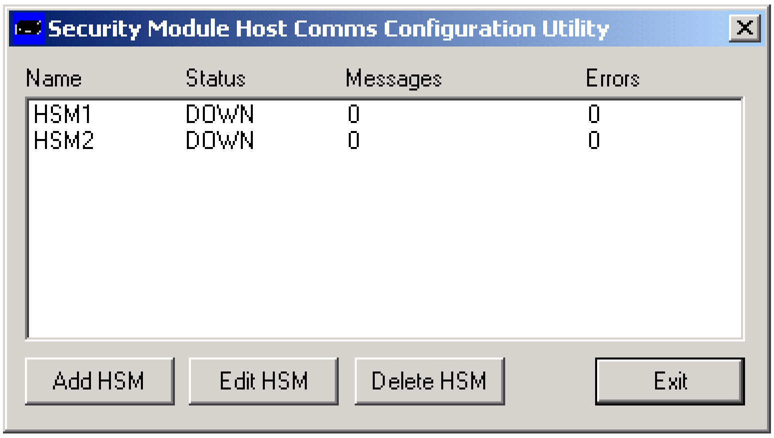
i	**Note**: The utility is contained in the ETptkcomm package.

For each HSM, you can configure the **Name**, **Weight**, **Window**, **Timeout**, and **Heartbeat**, and mark the **HSM Printer attached** and **Stand by only**.

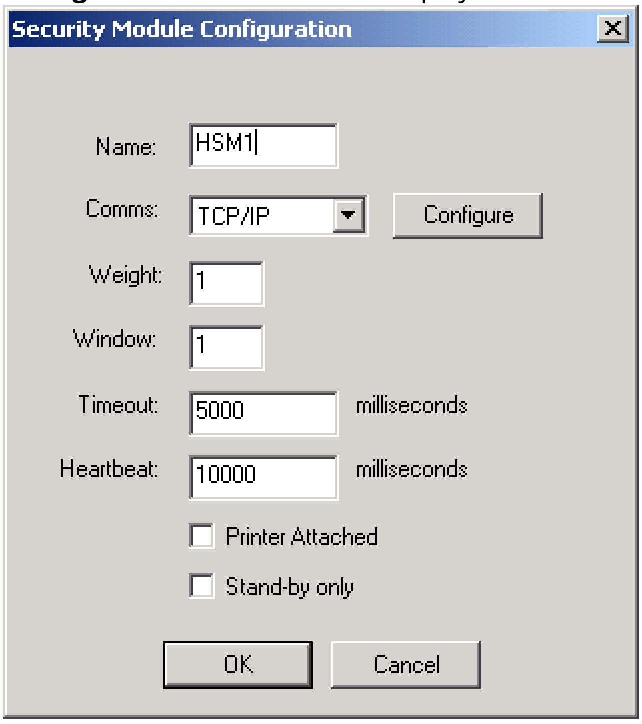
> **Notes**:
>
> - **Name** must be between one to four characters.
> - **Weight** specifies the distribution of the workload among HSM's. For example: If there are two HSM's, HSM1 with weight of four and HSM2 with weight of one, then HSM1 handles four functions before HSM2 handles one function.
>   - **Weight** must be greater than zero in order the HSM to be used. Weight of zero means the HSM is disabled or will not be used.
> - **Window** specifies the number of sent messages that can be outstanding for this security module before the receive queue is allowed time to catch up.
>   - If **Window** is greater than one, there is no guarantee that the responses will be returned in order.
> - **Timeout** is the amount of time to wait from sending the message before returning an error or retrying the send operation.
> - **Heartbeat** is the amount of time between sending heartbeat messages to the HSM. Heartbeat message is used to update the status of the HSM (UP or DOWN).
>   - **Heartbeat** should be greater than timeout.
> - **Printer attached** specifies that this HSM is connected to a printer for PIN or password printing.
> - **Stand by only** specifies that this HSM is only to be used when there is no other HSM available. For HSM with printing task only, this option must be set (checked).

For Windows GUI version of Security Module Configuration	1. In the AccessMatrix server, run Security Module Configuration. The **Security Module Host Comms Configuration Utility** window will be displayed, as shown in

below.



2. Click on the **Add HSM** button to add an HSM to the list. A **Security Module Configuration** window will be displayed as shown below.

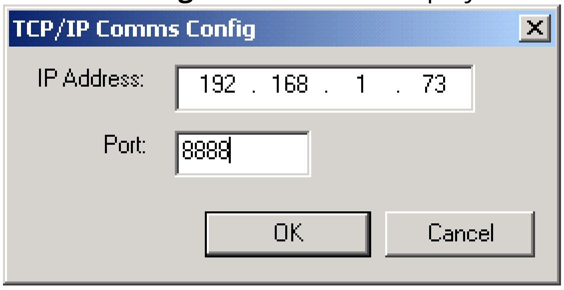


3. Enter the name, weight, window, timeout, heartbeat, printer attached, stand-by only information, accordingly.

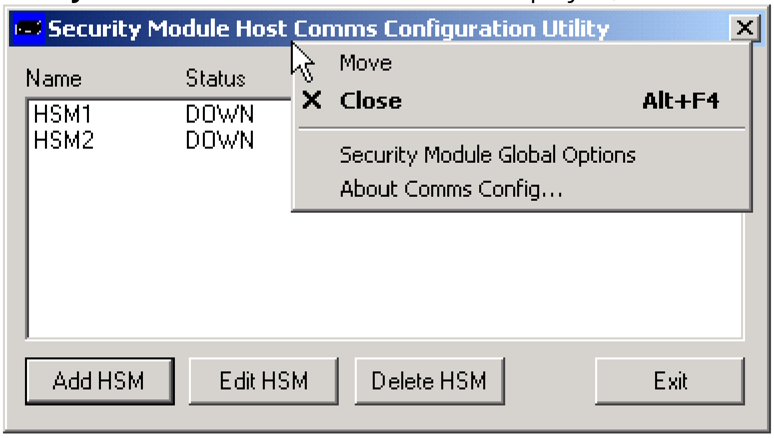
> **Notes**:
> - Please refer to the notes in the Configuring Security Module Host Comms section for more details.
> - HSM that handles printing task only, the Stand by only option must be set (checked) to prevent this HSM being used for password generation or verification.

4. Select **TCP/IP from Comms** to specify TCP/IP as the communication protocol used to communicate with the HSM.

5. Click on **Configure** to configure the TCP/IP communication with HSM. A **TCP/IP Comms Config** window will be displayed as shown below.



6. Enter the IP address and port of the HSM accordingly.
7. Click on the **OK** button to return to the Security Module Configuration window.
8. Click on the **OK** button to add the HSM to the list of security module and return to **Security Module Host Comms Configuration Utility** window.
9. Right click on the title bar of the **Security Module Host Comms Configuration Utility** window. A context menu will be displayed, as shown below.



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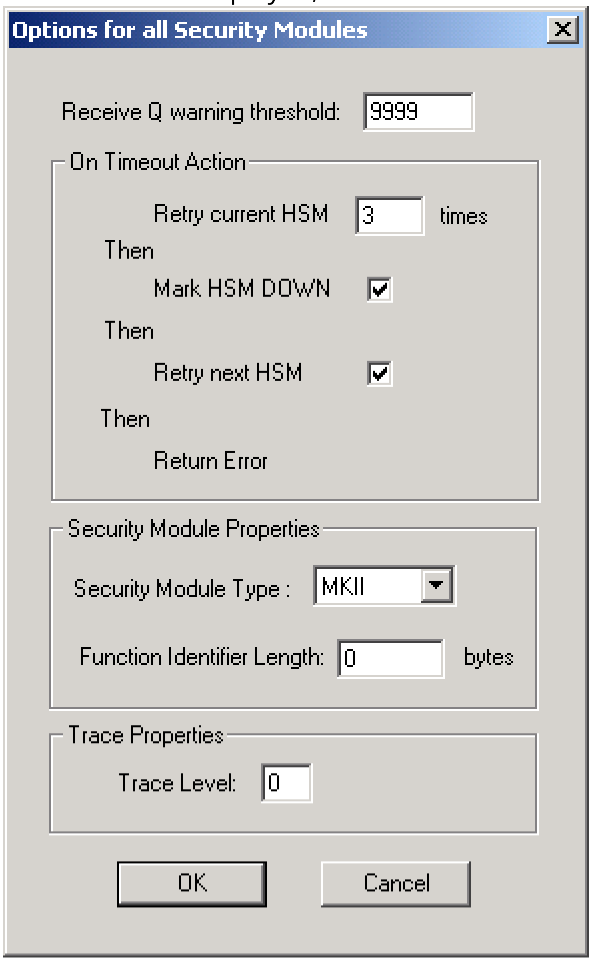
10. Click on Security Module Global Options. The Options for all Security Modules window will be displayed, as shown below.



11. Enter the number of retry current HSM and select the Mark HSM DOWN and Retry next HSM accordingly.

Notes:

Retry current HSM specifies how many times of retry in contacting the HSM after timeout error occurs.
If Mark HSM DOWN is checked, then the status of the timeout HSM will be marked as DOWN after certain number of retries.
Once HSM has been marked DOWN, the status of the HSM will be updated according to the heartbeat setting in step 3.
If Retry next HSM is checked, then the next step after marking the HSM DOWN is to retry using the next available HSM.
If all fails, then the timeout error will be returned to the caller.

12. Click on the OK button to save the configuration.
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<tr>
<td>For console version of Security Module Configuration</td>
<td>
Notes:

Default location of Eracom ETptkcomm installation in Solaris is /opt/ETptkcomm.
Default location of Eracom ETptkcomm installation in Windows is C:\Program Files\Eracom\ETptkcomm.

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</table>

	1. Locate and execute the commsconfig file (**commsconfig.exe** for Windows) in the ETptkcomm installation directory. The `Security Module Host Comms` configuration utility main menu will be displayed, as shown below. ``` Security Module Host Comms configuration utility Copyright (c) 2002 Eracom Technologies Version 4.0 Build 613 Runtime Version 4.0.613 *** Main Menu *** A - Add a new Security Module configuration E - Edit a Security Module configuration D - Delete a Security Module configuration P - edit global Security Module options ESC to exit ``` 2. Press the A button on the keyboard to add an HSM. The utility will ask for information about the new HSM configuration. ``` Enter Security Module name : HSM3 name = HSM3 Enter the weight for this Security Module : 1 Enter the window for this Security Module : 10 Enter the timeout (ms) for this Security Module : 3000 Enter the heartbeat interval (ms) for this Security Module : 6000 Is there printer attached to the Security Module [Y/N] ? : Y Is this Security Module marked as Standby [Y/N] ? : N >>Select Security Module connection>>>> 0 - TCP/IP 1 - ASYNC 2 - PCI ESC to cancel ENTER to accept default (*) Enter IP address : 192.168.1.20 Enter the port : 8888 ``` 3. Enter the name, weight, window, timeout, heartbeat, printer attached, stand-by only information, accordingly. **Note**: Please refer to the notes in the Configuring Security Module Host Comms section for more details. 4. Choose TCP/IP for the connection method for the new HSM by pressing "0". 5. Enter the IP address and port number of the HSM.

	6. From Main Menu, press P on the keyboard button to modify the Global Security Module Options.
	`Edit Globals`
	`Enter the Q Threshold : 8888`
	
	`>>Select action on Timeout>>>>`
	` 0 - Perform Retries`
	`*1 - Error Immediately`
	` ESC to cancel`
	` ENTER to accept default (*)`
	
	`Enter the number of times to retry the Security Module : 3`
	`Mark Security Module as DOWN on Timeout [Y/N] ? : Y`
	`Try Next Security Module [Y/N] ? : Y`
	
	`>>Select Security Module functionality>>>>`
	`*0 - MKII`
	` 1 - AMB`
	` ESC to cancel`
	7. Enter the global queue threshold.
	8. Choose the action taken after HSM timeout.
	9. Choose whether to mark the failure HSM DOWN or not after the timeout.
	Note: Once HSM has been marked DOWN, the status of the HSM will be updated according to the heartbeat setting in step 3.
	10. Choose whether to try the next HSM or not after marking the failure HSM DOWN.
	Note: To be able to set this setting, mark the failure HSM DOWN must be set.
	11. You will be asked for Security Module Functionality. You do not need to set this option, so press the ESC button on the keyboard. The Main Menu will be displayed again.

Failover Load Balancing

Failover load balancing allows two or more HSMs to share the work according to the failover load balancing configuration. It also provides higher reliability as other HSM's can take over the work of a failure HSM.

Notes:

- Security Module Configuration in ETptkcomm is required for configuring failover load balancing. Please install the ETptkcomm in AccessMatrix server before continuing.
- Printing task cannot be shared among several HSMs. Thus, only one HSM should handle the printing task.
- HSM that handles printing task only, the Stand by only option must be set (checked) to prevent this HSM being used for password generation or verification.

1. Backup the necessary keys, including RSA keys, from the main HSM. Please refer to the Backing Up and Restoring Using SmartCard and Storing or Backing up Keys to SmartCard sections.
2. Restore the necessary keys to other HSMs for failover load balancing. Please refer to the Retrieving or Restoring Keys from SmartCard section.
3. Add HSM in Security Module Host Comm Configuration utility. Please refer to the Configuring Security Module Host Comms section.

3.2. Configuring the Luna Payment HSM Printer

Printer Connection

For printing, SafeNet Luna Payment HSM uses an AUX2 port and can only use a serial port printer. Please refer to the Rear Panel for rear panel diagram. A serial printer is usually configured as DTE; thus, a cable or an adaptor that crosses between AUX2 port of HSM and the printer is usually required. SafeNet Luna Payment HSM provides the adaptor along with the HSM. The adaptor is called Pin mailer Null Modem.

Example of printer connection (may vary for some printers):

Printer			HSM	
	DTE		DTE	
RX	Pin 3	←	Pin 2	TX
RTS	Pin 4	→ Flow Control Busy →	Pin 5	CTS
DSR	Pin 6	← HSM Present ←	Pin 20	DTR
DTR	Pin 20	→ Printer Operable →	Pin 6	DSR

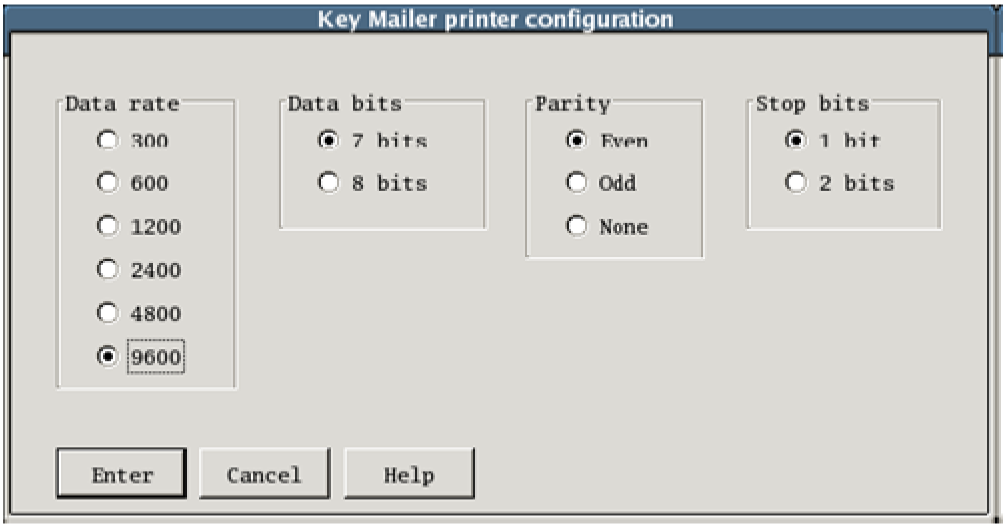
Configuring Printer Port

Note: Only serial port printer can be attached to HSM.

Printer must be properly configured for printing function. Please do the following to configure the printer port:

1. From the **Console Task Selection** menu, choose **OBM Pin Mailer** and press **Enter**. The **OBM Pin Mailer** menu appears.
2. From the **OBM Pin Mailer** menu, choose **OBM password mailer** and press **Enter**. The **OBM password mailer** menu appears.

3. From the **OBM password mailer** menu, choose **Printer port configuration** and press **Enter**. The **Key Mailer printer configuration** dialog appears, as shown below.



4. Choose the appropriate configuration for the attached printer.

> **Note**: The setting may vary from one printer to other printer. Please refer to the printer manual.

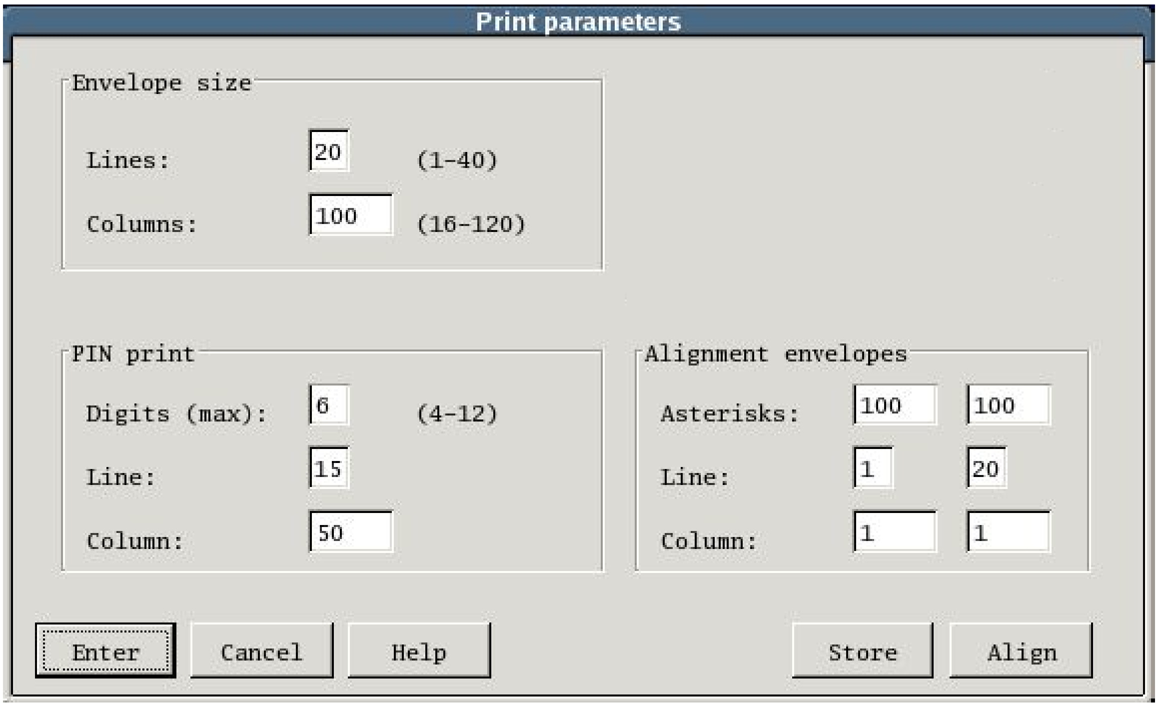
5. Select the **Enter** button.

Printing Alignment (Printing Parameters)

In order to print the password on envelope properly, the printing parameters must be properly set to align the printed password on the envelope:

1. From the **Console Task Selection** menu, choose **OBM Pin Mailer** and press **Enter**. The **OBM Pin Mailer** menu appears.
2. From the **OBM Pin Mailer menu**, choose **OBM password mailer** and press **Enter**. The **OBM password mailer** menu appears.

3. From the **OBM password mailer** menu, choose **Print parameters entry** and press **Enter**. The **Print parameters** dialog appears, as shown below.



4. Enter the size and location information:
   - **Envelope size** - **Lines**: Height of the envelope, measured in number of lines.
   - **Envelope size** - **Columns**: Width of the envelope, measured in number of characters.
   - **PIN print** - **Digits (max)**: Maximum number of characters of the password.
   - **PIN print** - **Line** and **Column**: Location of the password in the envelope. The password will be printed at the line and column specified in **Line** and **Column** respectively.
5. For alignment testing, you may enter up to two sets of asterisks, "*", to be printed:
   - **Alignment envelopes** - **Asterisks**: Number of asterisks, "*", to be printed for alignment test.
   - **Alignment envelopes** - **Line** and **Column**: Location to print the asterisks.
6. Please make sure the printer is ready.
7. Select the **Align** button to start printing the alignment test to the printer.
8. You may change the configuration and retest it until you are satisfied with the result.
9. You may choose from the following:
   - Select the **Enter** button to save the configuration temporarily. Once you leave PIN Mailer, the default setting will be retained.

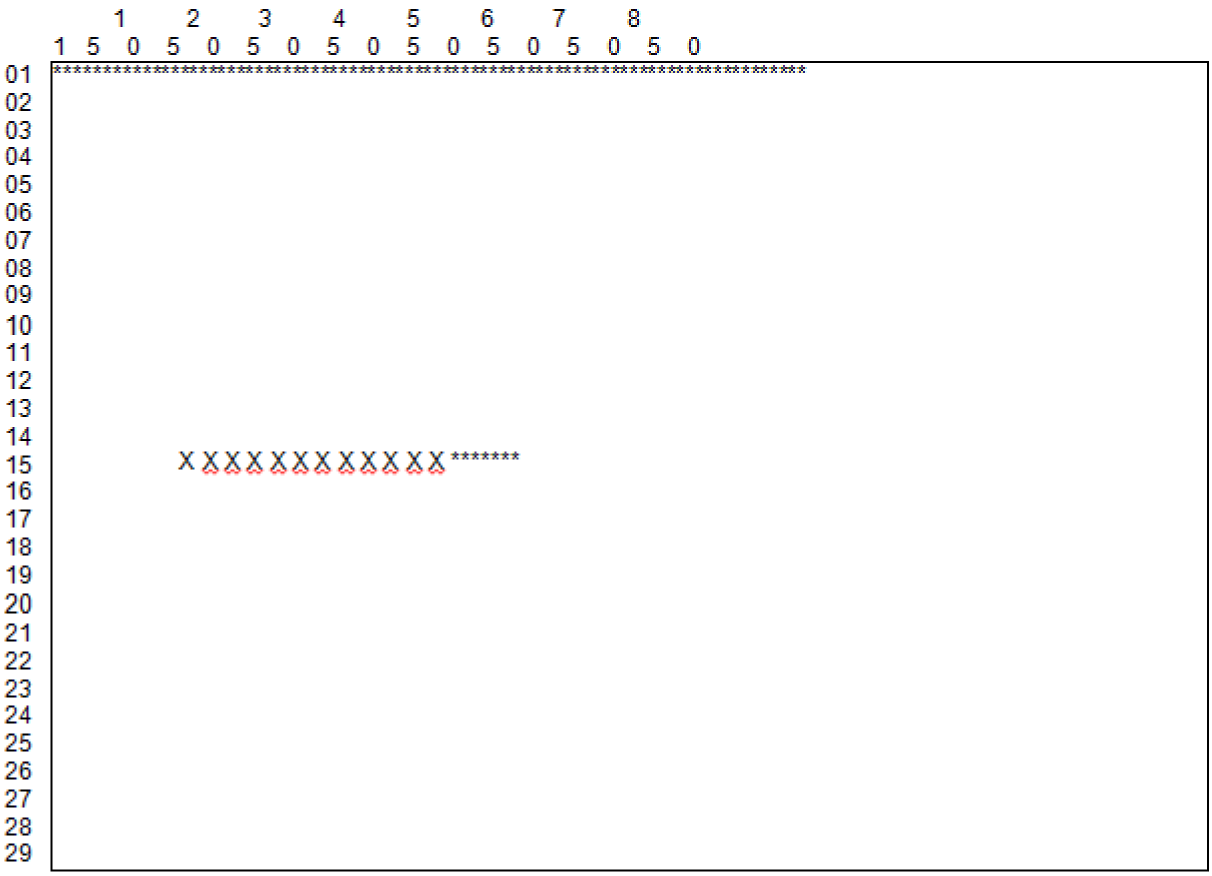
- Select the **Store** button to save the configuration permanently. The default setting will be replaced by this setting. This setting will be retained even after you leave PIN Mailer.

Printing Alignment Examples

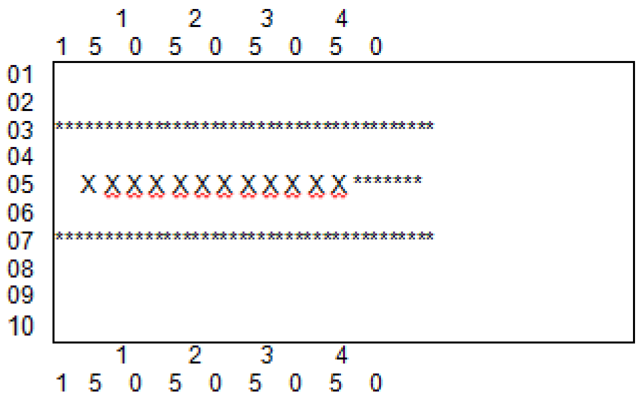
Please enter the following information for the print parameters:

Alignment Information	Example1		Example2	
Envelope size - Lines	30		10	
Envelope size - Columns	80		40	
Password print - Chars (max)	16		16	
Password print - Lines	15		5	
Password print - Columns	20		5	
Alignment envelopes - Asterisks	80	80	40	40
Alignment envelopes - Lines	1	30	3	7
Alignment envelopes - Columns	1	1	1	1

Below is the result of example 1. The box represents the envelope as specified in **Envelope size - Lines** and **Columns**. Asterisks (*) are printed according to the **Alignment envelopes - Asterisks**, **Lines**, and **Columns**.



Below is the result of example 2.



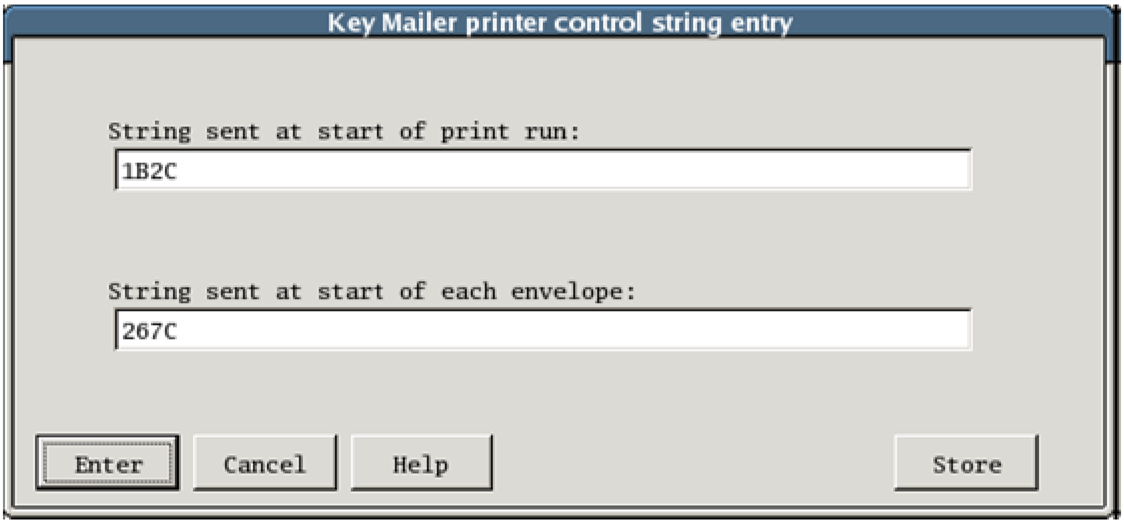
Printer Control Strings Setting

The **Key Mailer printer control string** dialog box allows for the entry of two printer control strings. The strings make it possible for the printer to be configured so that it is more suitable for a particular key mail envelope design. For example, the printer may be configured to print a more suitable number of lines per inch. Each control string consists of an even number of hexadecimal characters in the defined range of 01 to FF.

> **Note**: The control string entries have to be left justified and cannot contain embedded spaces.

1. From the **Console Task Selection** menu, choose **OBM Pin Mailer** and press **Enter**. The **OBM Pin Mailer** menu appears.
2. From the **OBM Pin Mailer** menu, choose **OBM password mailer** and press **Enter**. The O**BM password mailer** menu appears.

3. From the **OBM password mailer** menu, choose **Print control string entry** and press **Enter**. The **Key Mailer printer control string entry** dialog appears, as shown below.



4. Enter the control strings accordingly.

> **Notes**:
> - Please refer to your printer manual for the control string, as different printer has different control string.
> - String sent at start of print run is sent to printer before the first envelope and at the start of password mailer run.
> - String sent at start of each envelope is sent to printer before each envelope.

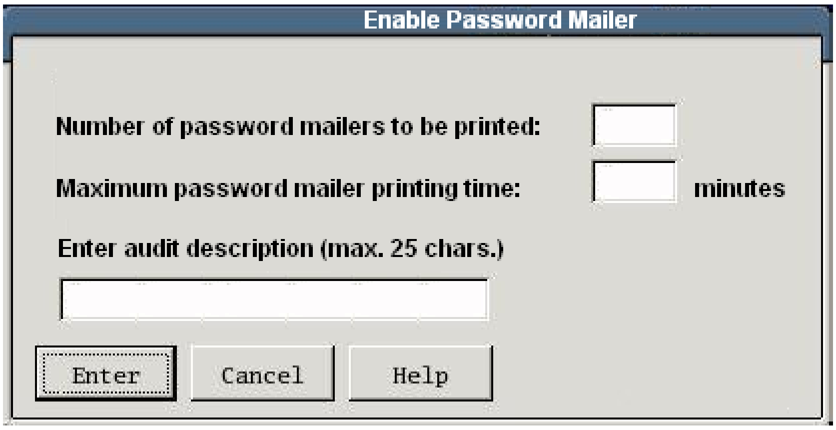
5. Choose from the following:
   - Select the **Enter** button to save the configuration temporarily. The setting will be reverted back to the default value when you leave the key mailer.
   - Select the **Store** button to save the configuration permanently. The setting will be saved as the default. Therefore, the setting will be retained even after restarting the HSM.

Enabling Password Mailer

Because of the sensitivity of passwords, a limit is imposed on the number of password mailers to be printed, and also on the time allowed for printing. After the limit has been exceeded, then the HSM will automatically turn off or disable the OBM Password Mailer Switch. To turn on the OBM Password Mailer Switch, please refer to the Enabling or Disabling OBM Password Mailer Switch section.

1. From the **Console Task Selection** menu, choose **OBM Pin Mailer** and press **Enter**. The **OBM Pin Mailer** menu appears.

2. From the **OBM Pin Mailer** menu, choose **OBM password mailer** and press **Enter**. The **OBM password mailer** menu appears.
3. From the **OBM password mailer** menu, choose **Password mailer enable** and press **Enter**. The **Enable password mailer** dialog appears, as shown below. To print alignment, please refer to the Printing Alignment (Printing Parameters) section.



4. Enter the following information:
    - **Number of password mailers to be printed** - After the HSM has printed the specified number of password mailers, the printing function will be disabled.
    - **Maximum password mailer printing time** - After the specified period, the password printing function will be disabled.
    - **Enter audit description** - This states the reason for enabling the password mailer. This information will be displayed in the audit report.
5. Select the **Enter** button. The **Printing in Progress** dialog will appear.

> **Notes**:
> - Leave the **Printing in Progress** dialog opened until the print run has finished.
> - Closing the dialog will disable the **OBM Password Mailer Switch**.

3.3. Key Processing

Key Processing Terminology

- **Dual Custody** - To assist in preserving key secrecy, at least two custodians are needed for all stored keys. Each has his own secret key component so neither custodian knows the whole key.
- **Clear Key Components** - A clear key component is a secret value known only to one custodian. Depending on the length of the key to which it belongs, it is a 16 or 32, hexadecimal digit, number. Each clear key component has a component number associated with it.
- **Encrypted Key Component** - An encrypted key component is similar to a clear key component, but it is treated differently by the key entry mechanism. It also has a component number associated with it.
- **The Resultant Key** - The key components described above are combined by the HSM to form a resultant key. This is the value stored in the HSM. The resultant key is never displayed.
- **Key Verification Codes (KVC)** - A KVC is used to verify, without compromising secrecy, that a key or a key component has been entered correctly, or to confirm that the value of a stored key is as expected. During key entry at the HSM console, a KVC is automatically calculated and displayed for each key component and for the resultant key. Additionally, facilities are provided to display the KVC's of the keys stored in each key table in secure memory.

Storing a Key

A common key storage system is used for all keys. Each key in the SafeNet Luna Payment HSM is uniquely identified by its type and its index. Each key type may have specific requirements.

When storing a key you need to know:

- The type of the key.
- The index of the key (its storage location).
- The number of clear key components it is comprised of.
- The number of encrypted key components it is comprised of.

A number of keys of each type can be stored at different index locations within the HSM. The number of index locations available depends on the key type.

Domain Master Keys (*KM)

Domain Master Key is used to encrypt other keys in the HSM or application passwords (e.g. database password). The SafeNet Luna Payment HSM can store three Domain Master Keys, defined as NEW_KM, CURRENT_KM, and OLD_KM. CURRENT_KM is the Domain Master Key that is currently used for encryption and decryption.

In order to change current Domain Master Key (CURRENT_KM), a new Domain Master Key (NEW_KM) must be entered. When a new Domain Master Key is entered, the key will be stored as NEW_KM. After that, the operator must establish the new Domain Master Key. This will move the CURRENT_KM to OLD_KM and NEW_KM to CURRENT_KM. Then a host function must be called to migrate the host-stored keys encrypted using the old Domain Master Key (OLD_KM) to the current one (CURRENT_KM). After all host-stored keys have been migrated, the OLD_KM can be deleted.

> **Notes**:
> - For creation of Domain Master Key, please refer to the Creating Domain Master Key (*KM) section.
> - For establish the new Domain Master Key (NEW_KM), please refer to the Establishing New Domain Master Key (NEW_KM) section.
> - To enable key migration for host-stored keys, the key migration must be enabled. Please refer to the Enabling/Disabling Key Migration for Domain Master Key section.
> - For deletion of old Domain Master Key (OLD_KM), please refer to the Deleting Old Domain Master Key (OLD_KM) section.

Creating Domain Master Key (*KM)

The key custodians may enter their keys for KTPV creation. If the key custodians would like their keys to be randomly generated, please refer to the Generating Random Keys or Components section.

> **Notes**:
> - The key length is always double-length.
> - There is no index number of Domain Master Key.

Please do the following to create a Domain Master Key:

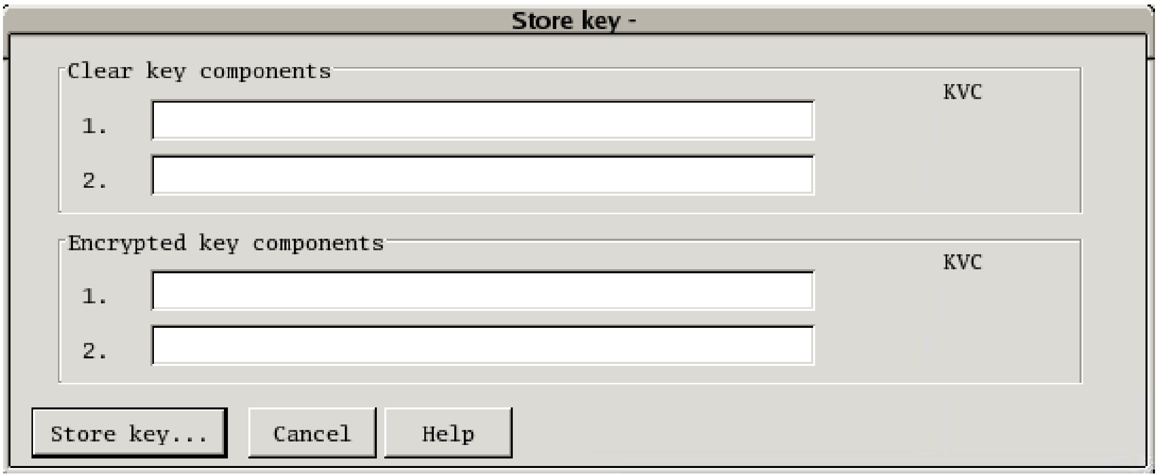
1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.

2. From the **Main Task Selection** menu, choose **Key processing** and press **Enter**. The **Key processing** menu appears.
3. From the **Key processing** menu, choose **Key entry** and press **Enter**. The **Key entry** menu appears.
4. From the **Key entry** menu, choose **Domain master keys (KM)** and press **Enter**. The **Domain master keys (KM)** menu appears.
5. If this is the first time of entering Domain Master Key (CURRENT_KM is empty), from the **Domain master keys (KM)** menu, choose **Store KM** and press **Enter**. The **Store key - KM** dialog appears as shown below.

6. If this is to store a new Domain Master Key (NEW_KM), from the **Domain master keys (KM)** menu, choose **Store new KM** and press **Enter**. The **Store key - KM dialog** appears as shown in previous step.
7. Enter the number of clear components and number of encrypted components.

> **Note**: At least two clear component parts are required.

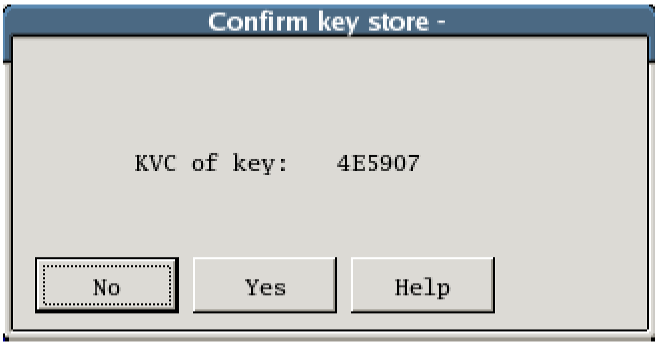
8. Select **Enter...** button. A dialog shown below will ask you for the clear key components and encrypted key components, if any.



9. Enter the clear key components and encrypted key components, if any.

> **Notes**:
> - The key components must be in hexadecimal (0-9A-F).
> - A KVC will be generated for confirmation.

10. Select the **Store key...** button. A key store confirmation will be displayed, as shown below.



11. Select the **Yes** button to create the KM.

Establishing New Domain Master Key (NEW_KM)

NEW_KM must be established to move the CURRENT_KM to OLD_KM and NEW_KM to CURRENT_KM.

Please do the following to establish new Domain Master Key (NEW_KM):

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Key processing** and press **Enter**. The **Key processing** menu appears.

3. From the **Key processing** menu, choose **Key migration** and press **Enter**. The **Key migration** menu appears.
4. From the **Key migration** menu, choose **Establish new domain master key** and press **Enter**. A status will be displayed.

Enabling/Disabling Key Migration for Domain Master Key

To allow migration of host-stored keys from OLD_KM to CURRENT_KM, the key migration must be enabled. The key migration should be disabled again after all host-stored keys had been migrated to CURRENT_KM.

Please do the following to enable or disable key migration:

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Key processing** and press **Enter**. The **Key processing** menu appears.
3. From the **Key processing** menu, choose **Key migration** and press **Enter**. The **Key migration** menu appears.
4. From the **Key migration** menu, choose **Enable/Disable key migration** and press **Enter**. The updated status will be displayed.

Deleting Old Domain Master Key (OLD_KM)

After all host-stored keys have been migrated to CURRENT_KM, the OLD_KM can be deleted.

Please do the following to delete old Domain Master Key (OLD_KM):

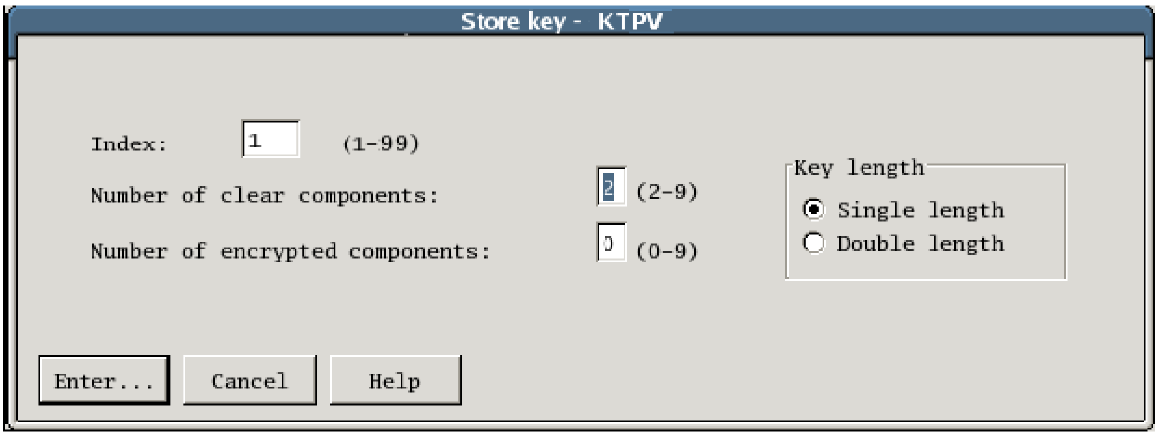
1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Key processing** and press **Enter**. The **Key processing** menu appears.
3. From the **Key processing** menu, choose **Key migration** and press **Enter**. The **Key migration** menu appears.
4. From the **Key migration** menu, choose **Enable/Disable key migration** and press **Enter**. The updated status will be displayed.

Creating Transformed PIN Value Key (KTPV)

KTPV is used to encrypt passwords. The key custodians may enter their keys for KTPV creation. If the key custodians would like their keys to be randomly generated, please refer to Generating Random Keys or Components section.

Please do the following to create a KTPV:

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Key processing** and press **Enter**. The **Key processing** menu appears.
3. From the **Key processing** menu, choose **Key entry** and press **Enter**. The **Key entry** menu appears.
4. From the **Key entry** menu, choose **Transformed PIN value key (KTPV)** and press **Enter**. The **Transformed PIN value key (KTPV)** menu appears.
5. From the **Transformed PIN value key (KTPV)** menu, choose **Enter KTPV** and press **Enter**. The **Store key - KTPV** dialog appears, as shown below.

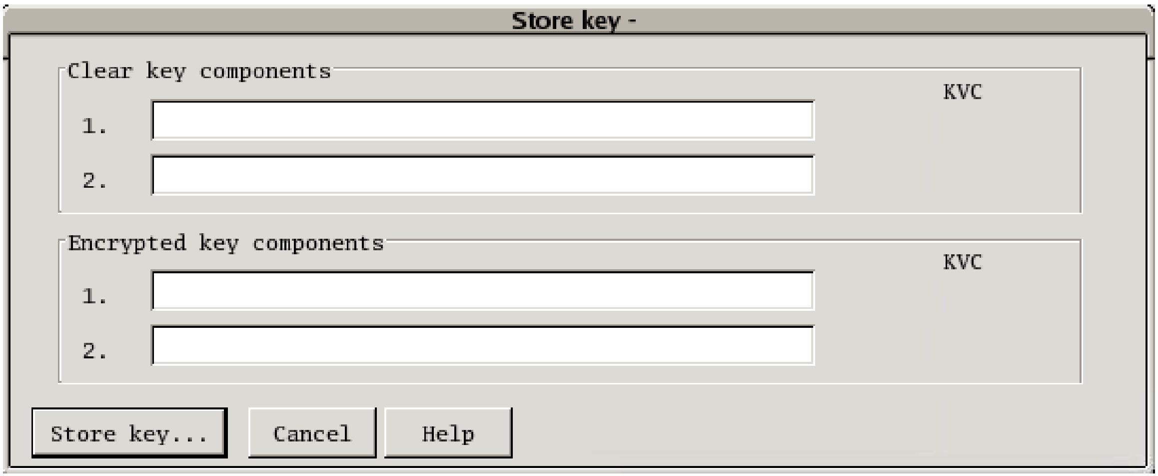


6. Enter the key index that you would like to add in the **Key Index** field.
7. Enter the number of clear components and number of encrypted components.

> **Note**: At least two clear component parts are required.

8. Select whether you want **Single Length** or **Double Length** for the **Key Length**.

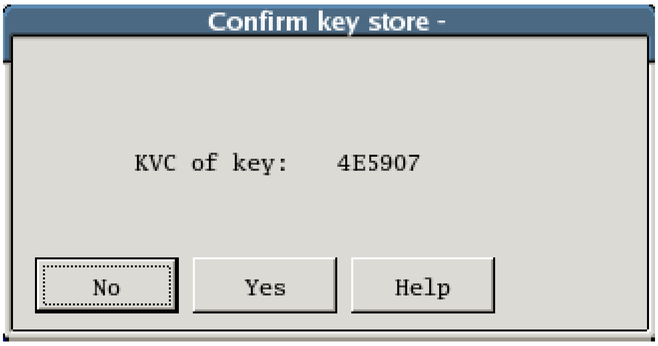
9. Select **Enter...** button. A dialog shown below will ask you for the clear key components and encrypted key components, if any.



10. Enter the clear key components and encrypted key components, if any.

> **Notes**:
> - The key components must be in hexadecimal (0-9A-F).
> - A KVC will be generated for confirmation.

11. Select **Store key...** button. A key store confirmation will be displayed, as shown below.



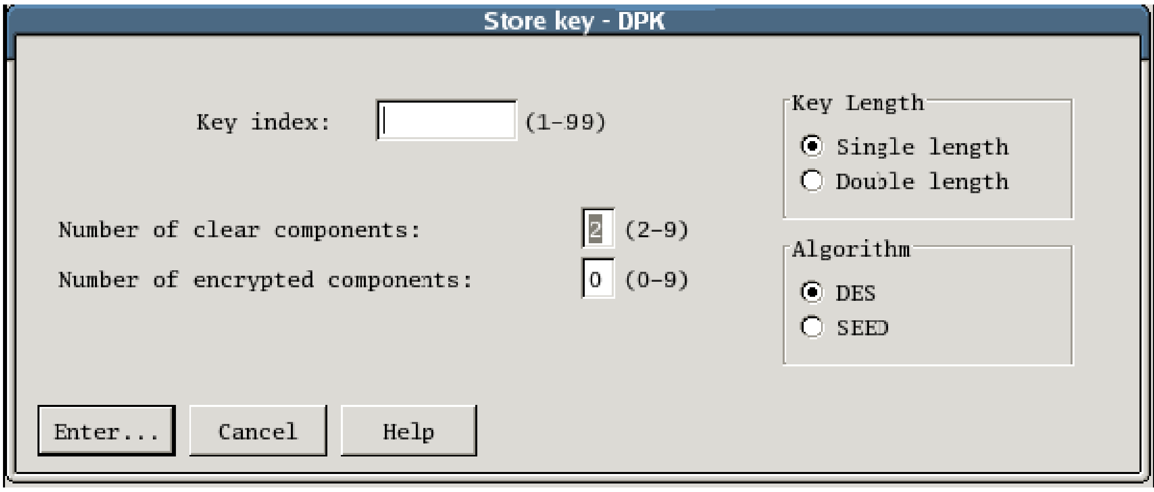
12. Select **Yes** button to create the KTPV.

Creating Data Protect Key (DPK)

Data Protect Key is used for encrypting various application's passwords, e.g. database password, application configuration password. The key custodians may enter their keys for DPK creation. If the key custodians would like their keys to be randomly generated, please refer to Generating Random Keys or Components section.

Please do the following to create a DPK:

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Key processing** and press **Enter**. The **Key processing** menu appears.
3. From the **Key processing** menu, choose **Key entry** and press **Enter**. The **Key entry** menu appears.
4. From the **Key entry** menu, choose **Data protect keys (DPK)** and press **Enter**. The **Data protect keys (DPK)** menu appears.
5. From the **Data protect keys (DPK)** menu, choose **Store DPK** and press **Enter**. The **Store key - DPK** dialog appears, as shown below.

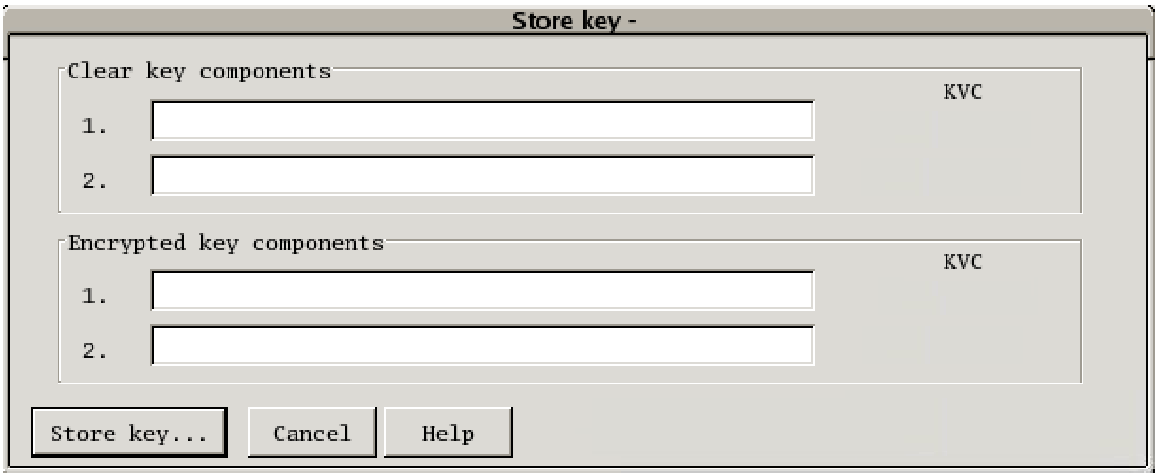


6. Enter the key index that you would like to add in the Key Index textbox.
7. Enter the number of clear components and number of encrypted components.

> **Note**: At least two clear component parts are required.

8. Select whether you want **Single length** or **Double length**.
9. Select the encryption algorithm.

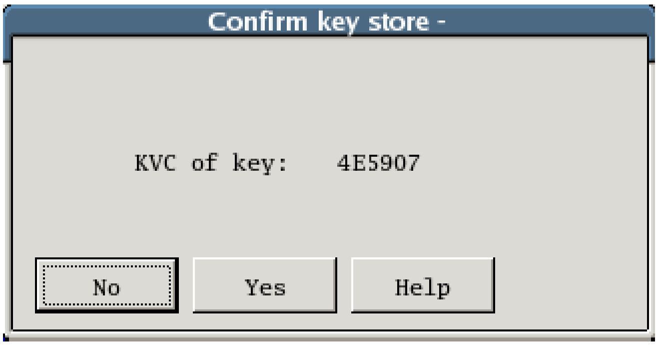
10. Select **Enter...** button. A dialog shown below will ask you for the clear key components and encrypted key components, if any.



11. Enter the clear key components and encrypted key components, if any.

Note: The key components must be in hexadecimal (0-9A-F).

12. A KVC will be generated for confirmation.
13. Select **Store key...** button. A key store confirmation will be displayed, as shown below.



14. Select the **Yes** button to create the DPK.

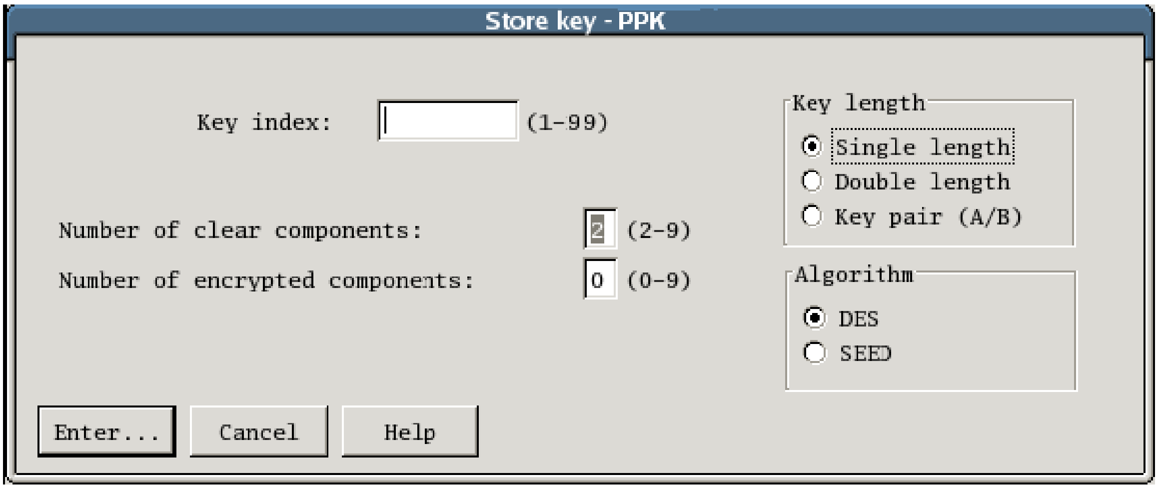
Creating PIN Protect Key (PPK)

PPK is used for encrypting/decrypting the password before sent for printing after being generated by PIN generation HSM. If the PIN generation and PIN printing functions are separated to two or more HSM's, then all of the HSM's must have the same PPK. For example, assume there are three HSMs: HSM A and HSM B for PIN generation, HSM C for PIN printing only. All of the HSMs must have the same PPK, so that after PIN generation by HSM A or HSM B, the generated PIN will be encrypted using the PPK before sending it to HSM C.

The key custodians may enter their keys for PPK creation. If the key custodians would like their keys to be randomly generated, please refer to Generating Random Keys or Components section.

Please do the following to create a PPK:

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Key processing** and press **Enter**. The **Key processing** menu appears.
3. From the **Key processing** menu, choose **Key entry** and press **Enter**. The **Key entry** menu appears.
4. From the **Key entry** menu, choose **PIN protect keys (PPK)** and press **Enter**. (Press **Page Up/ Page Down** to view **Next Page**). The **PIN protect keys (PPK)** menu appears.
5. From the **PIN protect keys (PPK)** menu, choose **Store PPK** and press **Enter**. The **Store key - PPK** dialog appears, as shown below.



6. Enter the key index that you would like to add in the **Key Index** field.
7. Enter the number of clear components and number of encrypted components.

> **Note**: At least two clear component parts are required.

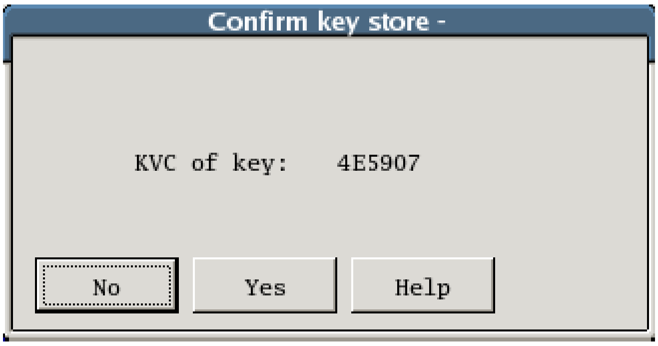
8. Select whether you want **Single Length**, **Double Length**, or **Key Pair (A/B)** for the **Key Length**.
9. Select the encryption algorithm.

10. Select the **Enter...** button. A dialog (shown below) will ask you for the clear key components and encrypted key components, if any.

11. Enter the clear key components and encrypted key components, if any.

> **Notes**:
> - The key components must be in hexadecimal (0-9A-F).
> - A KVC will be generated for confirmation.

12. Select the **Store key...** button. A key store confirmation will be displayed as shown below.



13. Select the **Yes** button to create the PPK.

Generating Random Keys or Components

SafeNet Luna Payment HSM can randomly generate keys or components for key entries, such as KTPV. Please follow the following steps for the random generation:

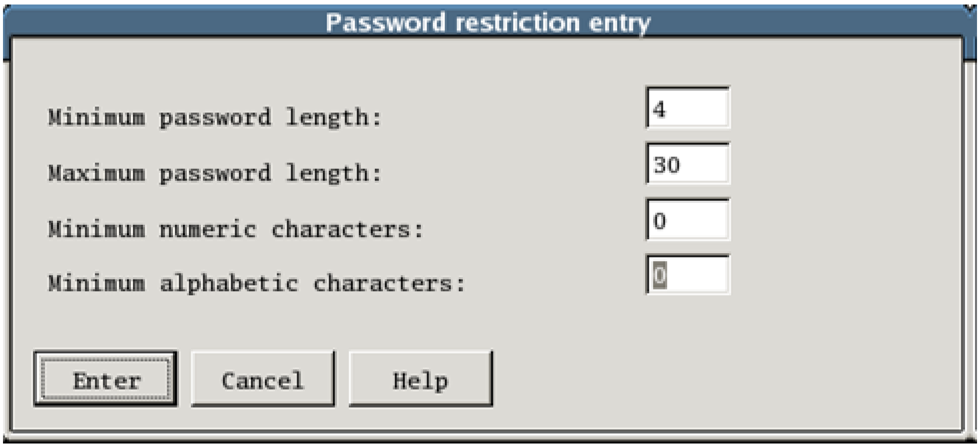
1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Key processing** and press **Enter**. The **Key processing** menu appears.

3. From the **Key processing** menu, choose **Display random keys/components** and press **Enter**. The **Random keys/components with verification codes** dialog appears showing the random generated keys and its KVC for both single and double length.
4. You may select the **Refresh** button to regenerate.

Setting Password Restriction

Password restriction setting is used to set the password policy for password printing, generation, and other password functions including set, change, and migrate. In order to set password policy for passwords that are sent to the HSM, please do the following:

1. From the **Console Task Selection** menu, choose **OBM Pin Mailer** and press **Enter**. The **OBM Pin Mailer** menu appears.
2. From the **OBM Pin Mailer** menu, choose **OBM password mailer** and press **Enter**. The **OBM password mailer** menu appears.
3. From the **OBM password mailer** menu, choose **Password restriction entry** and press **Enter**. The **Password restriction entry** dialog appears, as shown below.



4. Enter the restriction information accordingly.
5. Select the **Enter** button to save the restriction.

3.4. Backing Up and Restoring Using SmartCard

SafeNet Luna Payment HSM provides backup and restore functionalities to save and restore important keys stored in the HSM. The backup procedure can be done by transferring the data to smartcard. The data in the smartcard is encrypted using a Smartcard Transfer Protect Key (*KTP) and is protected by an Id and a PIN. Before restoring the data, the KTP must be entered into the HSM where the data is to be restored in order to decrypt the data. The Id and PIN are also required to restore the data.

> **Notes**:
>
> - By default, HSM RSA keys are not transferred to the smartcard. In order to enable the HSM RSA key transfer, please refer to Enabling Transferable HSM RSA keys section.
> - Backup and restore procedure can be used to copy the important keys to other HSM for failover load balancing, which is explained in Failover Load Balancing section.
> - The backup may require more than one smartcard to store the data, especially if HSM RSA keys are also to be transferred.
>
> **Warning:** The same KTP used to decrypt when restoring from the smartcard must be the same as the KTP used to encrypt the backup data to the smartcard. Thus, it is imperative that KTP must also be stored in a safe place for data retrieval.

SmartCard Transfer Protect Key (*KTP)

Before you can transfer keys to, or retrieve keys from the smartcard, you have to enter and store a double length Transfer Protect Key (*KTP) using the Store key - *KTP console operation. This key is used to derive double length keys to protect all data (including keys) in the smartcard transfer key and receive key process. The *KTP is entered in the standard manner as other SafeNet Luna Payment HSM keys.

> **Notes**:
>
> - Please refer to the Key Processing section for more information about key processing.
> - The same KTP used to decrypt when restoring from the smartcard must be the same as the KTP used to encrypt the backup data to the smartcard.

Please do the following to store *KTP:

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.

2. From the **Main Task Selection** menu, choose **Operations** and press **Enter**. The **Operations** menu appears.

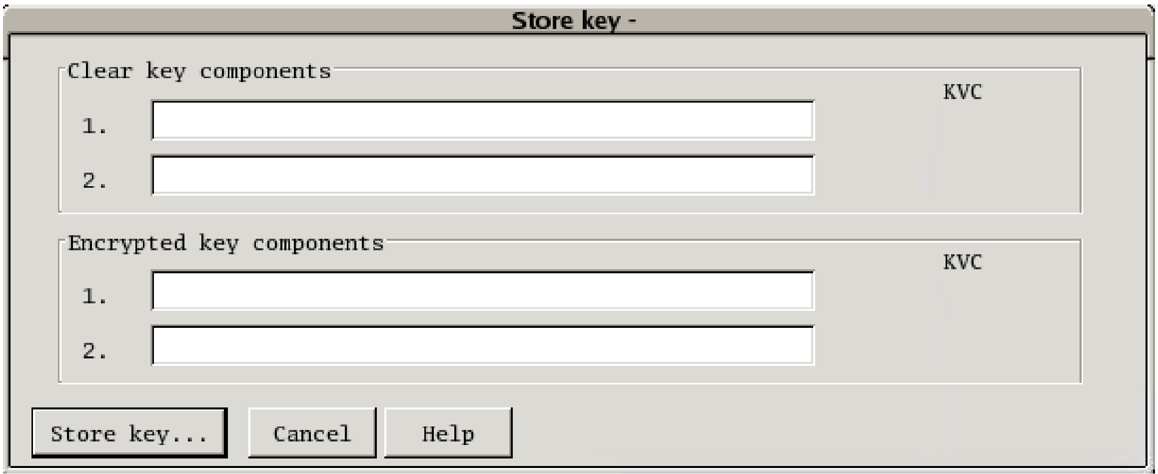
3. From the **Operations** menu, choose **Smartcard Key Transfer** and press **Enter**. The **Smartcard Key Transfer** menu appears.

4. From the **Smartcard Key Transfer** menu, choose **Store Transfer Protect Key (*KTP)** and press **Enter**. The **Store key - KTP** dialog appears as shown below.

5. Enter the number of clear components and number of encrypted components.

> **Note**: At least two clear component parts are required.

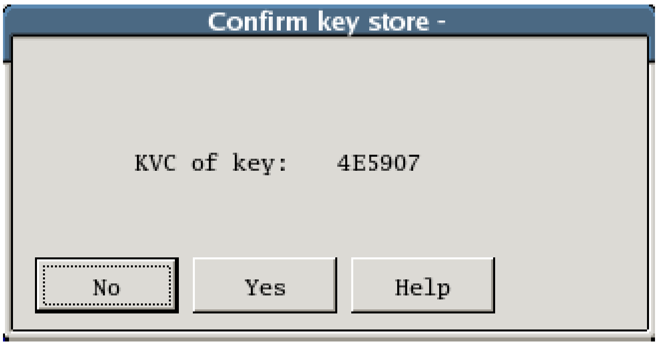
6. Select **Enter...** button. A dialog will ask you for the clear key components and encrypted key components, if any as shown below.



7. Enter the clear key components and encrypted key components, if any.

> **Notes**:
> - The key components must be in hexadecimal (0-9A-F).
> - A KVC will be generated for confirmation.

8. Select **Store key...** button. A key store confirmation will be displayed, as shown below:



9. Select the **Yes** button to create the KTP.

Storing or Backing up Keys to SmartCard

> **Note**: The same KTP used to decrypt when restoring from the smartcard must be the same as the KTP used to encrypt the backup data to the smartcard.

1. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
2. From the **Main Task Selection** menu, choose **Operations** and press **Enter**. The **Operations** menu appears.
3. From the **Operations** menu, choose **Smartcard Key Transfer** and press **Enter**. The **Smartcard Key Transfer** menu appears.
4. From the **Smartcard Key Transfer** menu, choose **Smartcard Transfer of all keys** and press **Enter**. The **Smartcard Transfer of all keys** menu appears.
5. From the **Smartcard Transfer** of all keys, choose **Transfer all stored keys to Smartcard set** and press **Enter**. The **Transfer all stored keys to SmartCards** dialog appears.
6. Set the **Set ID of the smartcard set** to the keys that are going to be written.
7. Select the **Enter** button. The **Smartcard PIN Entry** dialog appears.
8. Enter **Smartcard PIN** for the set.
9. Select the **Enter** button. The **Key Transfer in progress** dialog appears.
10. Once enough data has been read, insert the smartcard, as instructed. And select the **Enter** button. A notification dialog will appear to warn you that any existing data in the card will be destroyed.
11. Select the **OK** button to continue.

12. Follow the instruction on the screen.
13. The HSM will ask you to insert the first card again for writing the total number of cards used. Please do so.
14. A notification appears to let you know that the operation is successful. Select the **OK** button.

Retrieving or Restoring Keys from SmartCard

Notes:

- The same KTP used to decrypt when restoring from the smartcard must be the same as the KTP used to encrypt the backup data to the smartcard.
- Existing key in the HSM will be replaced if the existing key has the same index number as the one in the smartcard. However, if the keys in the smartcard do not use the existing key's index, then the existing key will not be deleted or replaced.
- If the smartcard contains RSA keys, the RSA keys will also be transferred or restored from the smartcard.

1. Insert the first smartcard of the smartcard set.
2. From the **Console Task Selection** menu, choose **Main Task Selection** and press **Enter**. The **Main Task Selection** menu appears.
3. From the **Main Task Selection** menu, choose **Operations** and press **Enter**. The **Operations** menu appears.
4. From the **Operations** menu, choose **Smartcard Key Transfer** and press **Enter**. The **Smartcard Key Transfer** menu appears.
5. From the **Smartcard Key Transfer** menu, choose **Smartcard Transfer of all keys** and press **Enter**. The **Smartcard Transfer of all keys** menu appears.
6. From the **Smartcard Transfer of all keys** menu, choose **Retrieve all stored keys from Smartcard set** and press **Enter**. The **Retrieve all stored keys from Smartcards** dialog appears.
7. Select the **Enter** button. A **Smartcard PIN Entry** dialog appears.
8. Enter the PIN number.
9. Select the **Enter** button. After verification, enter the **Set ID**.
10. Select the **Enter** button. After verification, the HSM will display how many bytes have been read from the card. Then the HSM will store the retrieved data into the Key Management System.
11. A notification will be displayed when the retrieval has been completed.

4. Appendices

This section contains the following appendices:

- A. Initial Setup Information
- B. Key Component Sample Form
- C. Erasing SafeNet Luna Payment HSM's Memory

4.1. A. Initial Setup Information

	Configuration	Attribute	Default/Recommended Value	Value
1.	Main Task Selection access menu password	New password		Do not write the password here
2.	SmartCard access menu password	New password		Do not write the password here
3.	OBM PIN Mailer access menu password	New password		Do not write the password here
4.	Date and time	Date	<current date>	
		Time	<current time>	
5.	Host port configuration	IP address of HSM		
		Subnet Mask	255.255.255.0	
		Service Port	8888	
		Frame Protocol	DIX-Ethernet	
		Max. Connects 1	8	
		Host Address (Client) 1		
		Gateway Address 1		
		Max. Connects 2	8	
		Host Address (Client) 2		
		Gateway Address 2		
		Max. Connects 3	8	
		Host Address (Client) 3		
		Gateway Address 3		

	Configuration	Attribute	Default/Recommended Value	Value
6.	Security Module Host Comms Configuration - Adding a Security Module or HSM	Name		
		Weight	1	
		Window	1	
		Timeout	5000	
		Heartbeat	10000	
		Printer Attached	Unchecked	
		Stand by only	Checked	
		IP Address		
		Port	8888	
7.	Security Module Host Comms Configuration - Security Module Global Options	Retry current HSM	3	
		Mark HSM DOWN	checked	
		Retry next HSM	checked	
8.	HSM RSA key length management	Key length	1024	
9.	Domain Master Key (KM)	Number of clear components	2	
		Number of encrypted components	0	
		Clear key components		You may write the key component on the key component form
		Encrypted key components	[N/A, if number of encrypted is 0]	
10.	SmartCard Transfer Key (KTP)	Number of clear components	2	

	Configuration	Attribute	Default/Recommended Value	Value
		Number of encrypted components	0	
		Clear key components		You may write the key component on the key component form
		Encrypted key components	[N/A, if number of encrypted is 0]	
11.	Retrieve all stored keys from Smartcards	PIN number		Do not write the password here
		Set ID		
12.	Transformed PIN Key (KTPV)	Key index		
		Number of clear components	2	
		Number of encrypted components	0	
		Key length	Double length	
		Clear key components		You may write the key component on the key component form
		Encrypted key components	[N/A, if number of encrypted is 0]	
13.	Data Protect Key (DPK)	Key index		
		Number of clear components	2	
		Number of encrypted components	0	
		Key length	Double length	

	Configuration	Attribute	Default/Recommended Value	Value
		Algorithm	DES	
		Clear key components		You may write the key component on the key component form
		Encrypted key components	[N/A, if number of encrypted is 0]	
14.	PIN Protect Key (PPK)	Key index		
		Number of clear components	2	
		Number of encrypted components	0	
		Key length	Double length	
		Algorithm	DES	
		Clear key components		You may write the key component on the key component form
		Encrypted key components	[N/A, if number of encrypted is 0]	
15.	Printer configuration	Data rate	9600	
		Data bits	8	
		Parity	None	
		Stop bits	1	
16.	Print parameters or alignment	Envelope size-Lines	30	
		Envelope size-Columns	80	
		Password print-Chars (max)	16	

	Configuration	Attribute	Default/Recommended Value	Value
		Password print-Lines	15	
		Password print-Columns	20	
		Alignment envelopes-Asterisks	80 & 80	
		Alignment envelopes-Lines	1 & 30	
		Alignment envelopes-Columns	1 & 1	
17.	Printer control string entry	String sent at start of print run	[blank]	
		String sent at start of each envelope	[blank]	
18.	Password restriction entry	Minimum password length	6	
		Maximum password length	10	
		Minimum numeric characters	1	
		Minimum alphabetic characters	1	
19.	Enable OBM Password Mailer		On	
20.	Enable password mailer	Number of password mailers to be printed		

	Configuration	Attribute	Default/Recommended Value	Value
		Maximum password mailer printing time		
		Enter audit description (max 25 chars.)		

4.2. B. Key Component Sample Forms

The following page will contain some key component sample form, which can be used for backup or safekeeping. However, key component is a very sensitive information. Thus, if key component is written to the form. The form must be kept safe.

The sample form can also be used as a template to create a more suitable form for your company and purposes.

Below are sample forms for each key component:

Domain Master Key (KM) Sample Form

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
Domain Master Key (KM) – Clear Component ___ of ___

Name of Key Custodian

KM Clear Component

Key Verification Check (KVC) for KM Clear Component

✂ cut here

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
Domain Master Key (KM) – Clear Component ___ of ___

Name of Key Custodian

KM Clear Component

Key Verification Check (KVC) for KM Clear Component

Transformed PIN Value Key (KTPV)

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
Transformed PIN Value Key (KTPV) – Index ___ – Clear Component ___ of ___

Name of Key Custodian

KTPV Clear Component

Key Verification Check (KVC) for KTPV Clear Component

✂ cut here

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
Transformed PIN Value Key (KTPV) – Index ___ – Clear Component ___ of ___

Name of Key Custodian

KTPV Clear Component

Key Verification Check (KVC) for KTPV Clear Component

SmartCard Transfer Protect Key (KTP)

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
SmartCard Transfer Protect Key (KTP) – Clear Component ___ of ___

Name of Key Custodian

KTP Clear Component

Key Verification Check (KVC) for KTP Clear Component

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✂ cut here

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
SmartCard Transfer Protect Key (KTP) – Clear Component ___ of ___

Name of Key Custodian

KTP Clear Component

Key Verification Check (KVC) for KTP Clear Component

Data Protect Key (DPK)

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
Data Protect Key (DPK) – Index ___ – Clear Component ___ of ___

Name of Key Custodian

DPK Clear Component

Key Verification Check (KVC) for DPK Clear Component

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Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
Data Protect Key (DPK) – Index ___ – Clear Component ___ of ___

Name of Key Custodian

PPK Clear Component

Key Verification Check (KVC) for DPK Clear Component

PIN Protect Key (PPK)

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
PIN Protect Key (PPK) – Index ___ – Clear Component ___ of ___

Name of Key Custodian

PPK Clear Component

Key Verification Check (KVC) for PPK Clear Component

✂ cut here

i-Sprint Innovations Pte Ltd
Blk 750D Lobby 1 Chai Chee Road
#08-01 TechnoPark @ Chai Chee
Singapore 469004

Telephone: (65) 6244-3900
Facsimile: (65) 6244-8900
Website: www.i-Sprint.com

Online Banking Module (OBM)
PIN Protect Key (PPK) – Index ___ – Clear Component ___ of ___

Name of Key Custodian

PPK Clear Component

Key Verification Check (KVC) for PPK Clear Component

4.3. C. Erasing SafeNet Luna Payment HSM's Memory

In order to erase the SafeNet Luna Payment HSM's memory, please follow the following step:

> **Important Note**: All data will be lost once erase memory is perform. All keys, passwords, and configurations will be erased.

1. Make sure the HSM is turned on.
2. Insert the Erase Memory key (red colour-coded) to the Erase Memory keyhole at the front HSM interface.
3. Turn the Erase Memory key clockwise as far as you can, and hold for five seconds before releasing it.
4. Turn the Power key to the Off position.
5. Turn the Power key in Reset position, and hold for five seconds.